

RASEL MAHMUD

DATA ANALYST

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL PROFILE

DTU Computer Science and Engineering graduate focused on data analysis, with experience in research data analysis and applied machine learning. I work best with complex, imperfect datasets: structuring them, validating evidence, comparing analytical approaches, and turning results into clear conclusions. I am targeting data analyst roles where data quality, analytical reasoning, and model evaluation matter.

TECHNICAL SKILLS

Programming & Analysis: Python, Pandas, NumPy, Excel, Jupyter Notebook

Machine Learning & Statistics: Scikit-learn, classification, logistic regression, KNN, cross-validation, model evaluation, statistical testing

Data Processing & Validation: Data cleaning, text extraction, regex, PDFMiner, CSV, JSON, manual validation, data quality

Visualization & Tools: Matplotlib, Process Mining, Git, GitHub

EDUCATION

MSc, Computer Science and Engineering | Technical University of Denmark (DTU) | 2023 - 2026

Relevant coursework: Introduction to Machine Learning and Data Mining; Computational Tools for Data Science; Social Data Analysis and Visualization; Process Mining

BSc, Computer Science and Engineering | Green University of Bangladesh | 2017 - 2020

Relevant coursework: Database Systems; Artificial Intelligence; Introduction to Statistics and Probability; Numerical Methods; Data Structures and Algorithms

MASTER'S THESIS - LARGE SCALE DOCUMENT ANALYTICS

Generative AI Declaration Analysis in DTU Master's Theses

- Built an end-to-end Python pipeline covering metadata preparation, PDF extraction, cleaning, candidate retrieval, classification, validation, and statistical analysis.
- Processed 943 analyzable theses and 935 candidate passages across multiple DTU departments and years.
- Designed rule-based classification logic using regex, context windows, negation handling, and ambiguity guards; compared outputs with an independent LLM classifier.
- Manually validated evidence against full document context and evaluated results using agreement analysis, confusion matrices, and statistical tests.

SELECTED PROJECTS

World Happiness Report - Predictive Modelling

- Prepared and analyzed structured data for supervised learning; compared Logistic Regression and K-Nearest Neighbors models.
- Applied nested cross-validation for model selection and performance estimation, then used McNemar testing to compare model predictions.
- Visualized results in Matplotlib and interpreted model performance and error patterns.

Apollo Mission Dataset - Process Mining Analysis

- Analyzed NASA Apollo 1-17 mission event data across five mission phases using process mining and the Process Mining Project Methodology (PM²).
- Mapped recurring mission flows and compared execution paths to identify timing variation, exceptional paths, and operational deviations in the event data.

Hoursly - Work Hours Analytics Application

- Built a workhour tracking application that calculates actual vs paid hours, balances, and custom pay periods; added historical summaries and PDF export to make payroll records and discrepancies easier to review.

ADDITIONAL PROFESSIONAL EXPERIENCE

Frontend Developer | BrixOptim | 2021 - 2023

Delivered frontend development work in a professional team environment, translating requirements into functional digital solutions.

LANGUAGES

English: Fluent | Danish: Learning